

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

# ENVIRONMENTAL SCIENCE

## Paper 1 (Generated practice paper - Set 08)

Assessed: The physical environment, Energy resources, Pollution, Research methods

**Time allowed: 3 hours****Materials**

For this paper you may use:

- a calculator.

**Instructions**

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

**Information**

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

**For Examiner's Use**

| Question | Mark |
|----------|------|
| 1        |      |
| 2        |      |
| 3        |      |
| 4        |      |
| 5        |      |
| 6        |      |
| 7        |      |
| 8        |      |
| 9        |      |
| 10       |      |
| 11       |      |
| TOTAL    |      |

**Answer all questions in the spaces provided.**

**0 1**

The Universal Soil Loss Equation (USLE) is  $A = R \times K \times LS \times C \times P$ . Table 1 gives factor values for a farm.

**Table 1**

| Factor                  | Description                                                                                         | Value        |
|-------------------------|-----------------------------------------------------------------------------------------------------|--------------|
| R                       | Rainfall erosivity for the farm                                                                     | 110          |
| K                       | Soil erodibility: loam                                                                              | 0.67         |
| LS                      | Slope length and gradient: 120 m, 6%                                                                | 1.35         |
| C (crop type x tillage) | Beans (0.50) with spring ploughing (0.90); or wheat (0.35) with mulch tillage (0.60)                | see question |
| P                       | Ploughing up and down slope 1.0; contour ploughing 0.50; strip cropping with contour ploughing 0.25 | see question |

**Table 2**

| Soil erosion class | Rate of soil loss / $\text{t ha}^{-1} \text{ yr}^{-1}$ |
|--------------------|--------------------------------------------------------|
| Low                | under 5                                                |
| Moderate           | 5-10                                                   |
| High               | 10-20                                                  |
| Very high          | 20-30                                                  |
| Severe             | over 30                                                |

**0 1**

**1**

Calculate the annual soil loss for beans with spring ploughing, ploughed up and down the slope. Give your answer to one decimal place.

**[2 marks]**

**Answer**

t/ha/yr

**0 1**

**2**

Calculate the annual soil loss for wheat with mulch tillage and strip cropping with contour ploughing.

**[2 marks]**

Answer  t/ha/yr

**0 1 . 3** The natural rate of soil formation on the farm is  $1 \text{ t ha}^{-1} \text{ yr}^{-1}$ . Use your answers and Table 2 to evaluate the sustainability of the two farming systems. **[3 marks]**

.....

.....

.....

.....

.....

.....

**0 1 . 4** Explain why the USLE gives only an estimate of soil loss. **[2 marks]**

.....

.....

.....

.....

**0 1 . 5** Suggest which single factor in the USLE a farmer can change most easily. **[1 mark]**

.....

.....

**10**

0 2 . 1

Explain how a ground-source heat pump can deliver more heat energy to a building than the electrical energy used to run it.

[3 marks]

.....

.....

.....

.....

.....

.....

0 2 . 2

Explain how Concentrating Solar Power (CSP) with thermal storage overcomes the intermittency of solar power.

[3 marks]

.....

.....

.....

.....

.....

.....

0 2 . 3

Explain why CSP can only be used in areas with little cloud, whereas PV can be used anywhere.

[2 marks]

.....

.....

.....

0 2 . 4

Describe **two** environmental impacts of solar PV.

[2 marks]

.....

.....

.....

1 .....

2 .....

10

03 . 1

Describe how bioleaching is used to extract copper from low-grade ores or spoil heaps, and explain why it may be more sustainable than smelting.

[5 marks]

03 . 2

Describe the process of phytomining.

[3 marks]

03 . 3

Explain how polymer adsorption may provide a low-energy source of uranium in the future.

[2 marks]

.....

.....

.....

.....

10

04 . 1

Explain how each of the following environmental factors affects the rate of degradation of pollutants. Give a named example of a pollutant for each factor.

[6 marks]

**Temperature** .....

.....

.....

.....

**Light** .....

.....

.....

.....

**Oxygen** .....

.....

.....

.....

04 . 2

Explain how the presence of adsorbent materials such as clay can reduce the severity of pollution by heavy metals or radioactive caesium-137.

[2 marks]

.....

.....

.....

.....

- 0 4 . 3** Explain why a fall in pH increases the harm caused by many heavy metals. **[2 marks]**

.....

.....

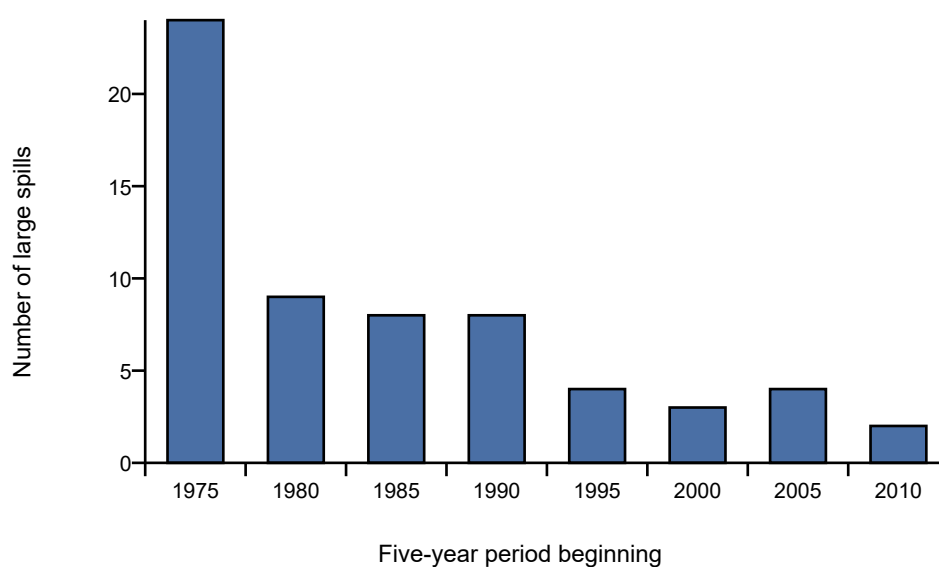
.....

.....

**10**

- 0 5** Figure 1 shows the number of large oil spills (over 700 tonnes) from tankers worldwide and the amount of oil transported by sea between 1975 and 2015.

**Figure 1**



**Table 1**

| Year                                    | 1975  | 1985  | 1995  | 2005  | 2015  |
|-----------------------------------------|-------|-------|-------|-------|-------|
| Oil transported by sea / million tonnes | 1 400 | 1 200 | 1 800 | 2 400 | 2 900 |

- 0 5 . 1** Calculate the percentage increase in oil transported by sea between 1975 and 2015. **[2 marks]**

Answer

%

05 . 2

Describe the trend in the number of large oil spills shown in Figure 1.

[2 marks]

.....

.....

.....

.....

05 . 3

Suggest **three** reasons for the decline in the number of large spills despite the increase in oil transported.

[3 marks]

1 .....

.....

2 .....

.....

3 .....

.....

05 . 4

Explain why the total amount of oil pollution entering the sea may not have fallen as much as the number of large tanker spills.

[3 marks]

.....

.....

.....

.....

.....

.....

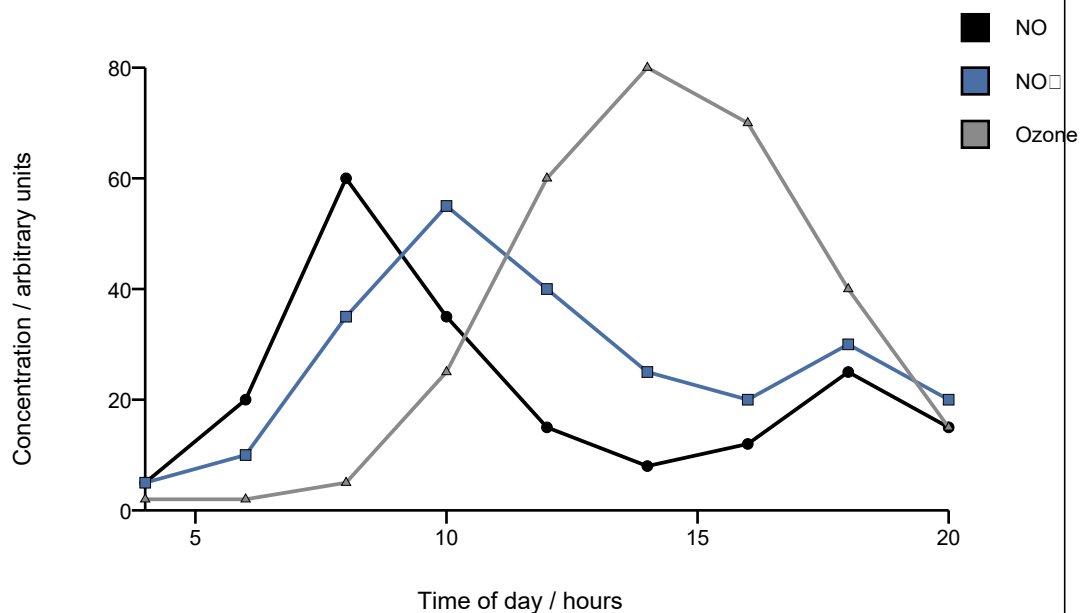
.....

10

0 6

Figure 1 shows the concentrations of three pollutants in the air of a large city in a valley during one sunny day.

Figure 1



0 6

1 Use Figure 1 to describe how the ozone concentration changes during the day and explain the pattern.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

0 6

2 Explain why the early-morning peak in NO is followed by a peak in NO<sub>2</sub>.

[2 marks]

.....

.....

0 6 . 3

Name **one** other toxic secondary pollutant produced in photochemical smogs and give one of its effects on humans.

[2 marks]

0 6 . 4

Explain why cities such as Los Angeles and Mexico City are particularly prone to photochemical smogs.

[2 marks]

10

0 7

A nuclear reprocessing plant on the coast discharges low-level liquid radioactive waste into the sea through a pipeline.

0 7 . 1

Explain what is meant by **Critical Pathway Analysis** (CPA).

[2 marks]

0 7 . 2

Give **three** factors that would be considered in a CPA for this discharge.

[3 marks]

1 .....

2 .....

3 .....

0 7 . 3

Describe how **Critical Group Monitoring** (CGM) would be used to assess the risk to the public from this discharge.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

0 7 . 4

Suggest **two** types of environmental sample that would be collected to check the CPA predictions.

[2 marks]

1 .....

2 .....

10

**0 8**

Figure 1 shows a simplified nitrogen cycle with some processes labelled 1 to 6.

**Figure 1**

| Number | Process description                                                            |
|--------|--------------------------------------------------------------------------------|
| 1      | Nitrogen gas converted to ammonium ions by Rhizobium in root nodules           |
| 2      | Ammonium ions oxidised to nitrite then nitrate by Nitrosomonas and Nitrobacter |
| 3      | Nitrates reduced to nitrogen gas by Pseudomonas in anaerobic soil              |
| 4      | Amino groups in dead organic matter released as ammonium ions by decomposers   |
| 5      | Nitrogen and oxygen react during lightning to form NO <sub>x</sub>             |
| 6      | Nitrates washed out of soil into rivers                                        |

**0 8****1**

Name processes 1 to 6.

**[3 marks]**

.....

.....

.....

.....

.....

.....

.....

**0 8****2**

Explain why draining a waterlogged field increases the nitrate available to crops.

**[3 marks]**

.....

.....

.....

.....

.....

.....

.....

08 . 3

Explain why nitrogen is described as a resource that is 'not very abundant' for living organisms even though the atmosphere is 78% nitrogen.

[2 marks]

08 . 4

Explain why growing legumes such as clover in a crop rotation reduces the need for artificial fertiliser.

[2 marks]

10

09 . 1

A student measured the pH of a stream with a meter that reads to 0.01 pH units but had not been calibrated. Which statement is correct? Tick **one** box.

[1 mark]

- A The readings are precise but may not be accurate
- B The readings are accurate but not precise
- C The readings are both accurate and precise
- D The readings are reliable because the meter is electronic

☐☐☐☐

09 . 2

Explain what is meant by an **anomalous result** and how a student should decide whether to include it.

[2 marks]

09 . 3

Explain why a preliminary study is carried out before the main investigation.

[2 marks]

5

10 . 1

Explain why the development of energy storage technologies is essential if renewable energy resources are to provide most of a country's electricity.

**[9 marks]**

Extra space

10 . 2 State what is meant by **peak shaving**.

[1 mark]

10

Do not write outside the box

**Write an essay on one of the following topics.**

|   |   |   |   |
|---|---|---|---|
| 1 | 1 | . | 1 |
|---|---|---|---|

Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

**[25 marks]**

**OR**

**1 1 . 2**

Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.







**END OF QUESTIONS**